

Udbhav Ram

Medical Physics · Clinical AI · Software

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PUBLIC CV · UPDATED OCTOBER 2026

Education

Ph.D. student, Medical Physics | 2026-Present

University of Wisconsin-Madison, Madison, WI

- Research interests include artificial intelligence for radiation oncology, image segmentation, vision-language models, treatment planning and image-guided radiotherapy.

B.Sc. Honours Medical Physics (with Co-op) | 2021-2026

McMaster University, Hamilton, ON

- Coursework included clinical applications of physics in medicine, radiation biology, and radiation and radioisotope methodology.

Research Experience

Research Assistant, Medical Physics / Radiology | June 2026-Present

University of Wisconsin-Madison, Madison, WI

- Graduate research with mentor Dr. Ran Zhang, with a focus on vision-language models and AI applications in medical imaging and radiation oncology.

International Visiting Scholar and Research Collaborator

Department of Radiation Oncology, University of Alabama at Birmingham, Birmingham, AL

Research collaboration began in 2021; visiting-scholar appointments in 2024 and 2025.

- Evaluated Ethos 2.0 high-fidelity mode for multi-metastasis, single-isocenter stereotactic radiotherapy using semi-automated comparative analysis.
- Investigated inter-observer contouring variability in adaptive accelerated partial breast irradiation and LLM-assisted TG-263 target-name quality assurance.
- Evaluated deep-learning frameworks for abdominal organ auto-segmentation and investigated machine-learning approaches to radiosensitivity and fractionated dose delivery.
- Completed 400 hours of clinical shadowing in radiation oncology and medical physics.

Undergraduate Thesis Student | Sept 2024-May 2025

Juravinski Cancer Centre, Hamilton, ON

- Conducted a JCC-UAB commissioning study comparing 6X and 10X flattening filter-free photon beams on a TrueBeam iTX linac for SBRT using the Eclipse treatment planning system.

Researcher, Department of Medical Physics | Aug 2022-Jan 2023

University of Western Ontario, London, ON

- Built deep-learning models for real-time optical-scan analysis of cerebral hemoglobin oxygenation.

Clinical Research Fellow, Department of Nephrology | Sept 2021-Aug 2022

St. Joseph's Hospital, Hamilton, ON

- Designed deep-learning and computer-vision tools for augmented-reality support in 5/6 nephrectomy mouse models.

Research Experience (continued)

High School Research Fellow, Physics and Astronomy | Sept 2019-Aug 2022

McMaster University, Hamilton, ON

- Investigated resveratrol, caffeine, beta-carotene and EGCG effects on amyloid aggregation in synthetic brain membranes.

Research Assistant, W Booth School of Engineering Practice and Technology | Jan-June 2023

McMaster University, Hamilton, ON

- Built and validated full-stack applications using JavaScript frameworks for academic transcript generation.

Professional & Work Experience

Data and Strategy Intern | June-Aug 2023

McLaren Racing, Arrow McLaren IndyCar, Indianapolis, IN

- Developed modelling pipelines, stochastic simulators and deep-learning computer-vision applications to support race, performance and data engineers.

Lead, Data Analytics and Strategy | Oct 2021-Jan 2024

MAC Formula Electric, Hamilton, ON

- Developed CAN communication protocols, vehicle-control integration and driver-dashboard systems.

Full Stack Engineer | 2020-2021

Syth-Med Biotechnology

- Developed full-stack applications for biotechnology solutions.

DevOps Technical Lead | 2020-2021

WaaW Global Inc.

- Led DevOps infrastructure and technical operations.

Ambassador | 2022-2023

Journal of Science Policy & Governance

- Promoted student engagement in science policy and governance.

Teaching & Mentoring Experience

Undergraduate Teaching Assistant, Physics and Astronomy | Sept 2024-May 2025

McMaster University, Hamilton, ON

- Supported undergraduate physics instruction in introductory mechanics and electricity and magnetism.

Mentor | Sept 2024-May 2025

McMaster Undergraduate Physics Society

- Helped physics students navigate courses, internships and extracurricular opportunities.

Mentor | Sept 2024-May 2025

McMaster Science Careers and Experience Centre

- Supported medical and biological physics co-op students with placement searches, application documents and interview preparation.

Head Yoga Instructor

McMaster University Athletics

- Led yoga instruction for student cohorts; role began in May 2022.

Peer-Reviewed Publications

Yogesh Kumar, Kayla L. Steed, Ho-Hsin Chang, Udbhav Ram, Gaurav N. Rathi, Katherine A. Heinzman, Ritish Nedunoori, Farnaz Rahim Li, Richard A. Popple, Rex Cardan, Christopher D. Willey, Drexell H. Boggs, John B. Fiveash, Carlos E. Cardenas. **AI-based framework to fuse pre-RT brain metastases contours with follow-up MRI to improve post-RT assessment.** *Neuro-Oncology Practice*, 13(3), 497–505. 2026-06. doi:10.1093/nop/npaf126

Udbhav S. Ram, Richard A. Popple, John B. Fiveash, Katherine A. Heinzman, Yogesh Kumar, Natalie N. Viscariello, Courtney B. Stanley, Dennis N. Stanley, Carlos E. Cardenas, Joel A. Pogue. **Dosimetric evaluation of Ethos 2.0 high-fidelity mode for single-isocenter SRS.** *Journal of Applied Clinical Medical Physics*, 26(12), e70370. 2025-12. doi:10.1002/acm2.70370

Udbhav S. Ram, Joel A. Pogue, Michael Soike, Neil T. Pfister, Rojymon Jacob, Carlos E. Cardenas. **Assessing quantitative performance and expert review of multiple deep learning-based frameworks for computed tomography-based abdominal organ auto-segmentation.** *Intelligent Oncology*, 1(2), 160–171. 2025-04. doi:10.1016/j.intonc.2025.03.003

Isabella P. Gastaldo, Sebastian Himbert, Udbhav Ram, Maikel C. Rheinstädter. **The Effects of Resveratrol, Caffeine, β -Carotene, and Epigallocatechin Gallate (EGCG) on Amyloid- β 25–35 Aggregation in Synthetic Brain Membranes.** *Molecular Nutrition & Food Research*, 64(22), e2000632. 2020. doi:10.1002/mnfr.202000632

Conference Contributions, Talks & Remarks

26 discrete event and remarks records, including one scheduled talk whose delivery is unverified, plus a separate outreach aggregate. Reconciled through October 6, 2026; not an exhaustive lifetime record.

Characterizing Contouring Uncertainty across Clinical Specialties in CBCT-Guided Online Adaptive Partial Breast Irradiation

September 29, 2026 · ASTRO 68th Annual Meeting

poster · Boston, Massachusetts, USA

first author and listed presenter · Presenter: Udbhav Ram, BS

Presented

3104

Event record and sources

Retrospective Weekly Photon Re-Optimization on QA Scans to Assess Convergence toward Proton Dosimetry during Treatment

September 29, 2026 · ASTRO 68th Annual Meeting

poster · Boston, Massachusetts, USA

Third author; coauthor, not named presenter · Presenter: Yogesh Kumar, PhD

Conference program record

3044

Event record and sources

Inter-Observer Variability of Clinical Target Volume Delineation In Cone Beam CT-Guided Online Adaptive Accelerated Partial Breast Irradiation

July 19, 2026 – July 22, 2026 · 2026 Joint AAPM | COMP Annual Meeting & Exhibition

digital-only poster · Online, within Joint AAPM | COMP 2026 (Vancouver meeting)

presenting author · Presenter: Udbhav S. Ram

Published digital poster

PO-DPV-T-14

Meeting dates; digital-only poster with no dedicated physical session

Event record and sources

Watch Once, Work Forever: One-Shot Computer-Use Agents for Clinical UI Automation

July 19, 2026 · 2026 Joint AAPM | COMP Annual Meeting & Exhibition

poster discussion · Vancouver, British Columbia, Canada

presenting author · Presenter: Udbhav S. Ram

Conference program record

SU-315-GPD-T-614

exact program viewing/discussion date

Event record and sources

Predicting Adaptive Radiotherapy Session Duration Using Reference Plan Calculation Metrics

July 19, 2026 · 2026 Joint AAPM | COMP Annual Meeting & Exhibition

poster discussion · Vancouver, British Columbia, Canada

coauthor · Presenter: Dennis N. Stanley, PhD

Conference program record

SU-315-GPD-T-501

exact program viewing/discussion date

Event record and sources

When APIs Don't Exist: Visual-Language-Driven Automation of Streamed Medical Software

July 19, 2026 · 2026 Joint AAPM | COMP Annual Meeting & Exhibition

poster discussion · Vancouver, British Columbia, Canada

coauthor · Presenter: Rex A. Cardan, PhD

Conference program record

SU-315-GPD-T-16

exact program viewing/discussion date

Event record and sources

The State of AI in Radiation Oncology and Medical Physics in North America: A Researcher's Perspective

May 14, 2026 · AIIMS New Delhi

Research seminar · New Delhi, India; attendance format unconfirmed

Scheduled speaker · Presenter: Udbhav Ram

Scheduled talk; delivery not independently verified

Event record and sources

Evaluation of Ethos 2.0 High-Fidelity Mode for Multi-Met, Single-Isocenter Stereotactic Radiotherapy: A Semi-Automated Comparative Analysis

October 31, 2025 · 2025 Physics and Astronomy Congress

Invited poster presentation · Denver, Colorado, USA

Invited poster presenter · Presenter: Udbhav Ram, University of Alabama at Birmingham

Presented

Session

Event record and sources

Dosimetric Comparison of 6X and 10X Flattening Filter-Free Photon Beams on a Truebeam Linac Using the Eclipse TPS

July 27, 2025 – July 30, 2025 · AAPM 67th Annual Meeting & Exhibition

digital-only poster · Online, within AAPM 2025 (Washington, DC meeting)

presenting author · Presenter: Udbhav S. Ram

Presented

PO-GPV-T-244

Meeting date range; digital poster, no physical session

Event record and sources

Improving TG-263 Target Name Compliance Using Locally-Hosted Large Language Models

July 27, 2025 · AAPM 67th Annual Meeting & Exhibition

Blue Ribbon Poster · Washington, DC, USA

presenting author · Presenter: Udbhav S. Ram

Presented

SU-330-BRP-12

exact program viewing/discussion date

Blue Ribbon Poster, AAPM 2025

Event record and sources

Evaluation of High-Fidelity Mode for Semi-Automated Multi-Met, Single-Isocenter Stereotactic Radiosurgery Planning Using the Ethos 2.0 Planning System

March 20, 2025 · 2025 Radiosurgery Society Scientific Meeting

Poster presentation · Tucson, Arizona, United States

First author; coauthored presentation · Presenter: Carlos E. Cardenas, PhD

Conference program record

Event record and sources

Dosimetric Comparison of 6MV and 10MV Flattening Filter-Free Photon Beams

2025 · Undergraduate thesis talk

Thesis research talk · McMaster University / Juravinski Cancer Centre; exact venue unspecified

Presenter · Presenter: Udbhav Ram

Recorded thesis presentation

Year confirmed; exact delivery day unknown

Event record and sources

McMaster UAB Partnership Summary

December 13, 2024 · McMaster/UAB student exchange planning meeting

Institutional/ambassador presentation · Location and attendance format unconfirmed

Presenter and student exchange advocate · Presenter: Udbhav Ram

Presented

Event record and sources

Evaluation of Ethos 2.0 High-Fidelity Mode for Multi-Met, Single-Isocenter Stereotactic Radiotherapy: A Semi-Automated Comparative Analysis

July 23, 2024 · AAPM 66th Annual Meeting & Exhibition

poster discussion · Los Angeles, California, USA

presenting author · Presenter: Udbhav S. Ram

Presented

TU300-GPD(B)-LOUNGE-149

exact program viewing/discussion date

Outstanding Poster Presentation, 2024 Society of Physics Students-AAPM undergraduate research competition

Event record and sources

Head-to-Head Comparison of State-of-the-Art Automl Segmentation Frameworks Using Publicly Available Data

July 23, 2024 · AAPM 66th Annual Meeting & Exhibition

poster discussion · Los Angeles, California, USA

presenting author · Presenter: Udbhav S. Ram

Presented

TU300-GPD(B)-LOUNGE-559

exact program viewing/discussion date

Event record and sources

Improving TG-263 target name compliance using locally-hosted large language models

June 5, 2024 · COMP Annual Scientific Meeting

Rapid oral presentation · Regina, Saskatchewan, Canada

Accepted rapid-presentation participant and presenter · Presenter: Udbhav Ram

Presented

Event record and sources

Evaluation of Ethos 2.0 High-Fidelity Mode for Multi-Met, single-isocenter stereotactic radiotherapy: A semi-automated comparative analysis

June 5, 2024 · 70th COMP Annual Scientific Meeting

Oral presentation in Science Spoken Succinctly (S3) research communication competition · Regina, Saskatchewan, Canada

S3 competition presenter · Presenter: Udbhav Ram

Conference program record

Session 87

Event record and sources

Student remarks at the UAB-McMaster partnership signing

2024 (approximate) · Inaugural UAB-McMaster memorandum of understanding signing meeting

Invited institutional student remarks / impromptu speech · Birmingham, Alabama, United States

Student representative and speaker · Presenter: Udbhav Ram

Retrospective institutional account

Year inferred from institutional retrospective; exact date unresolved

Event record and sources

Optimizing dose delivery during fractionated radiotherapy

October 28, 2023 · Canadian Undergraduate Physics Conference 2023

Contributed undergraduate oral presentation; first-place oral award · Waterloo, Ontario, Canada

Named presenter and first-place oral award recipient · Presenter: Udbhav Ram

Presented

First place, oral presentations, CUPC 2023

Event record and sources

Assessing Efficacy of Deep Learning-Based Auto-Contouring for Abdominal Normal Tissues

July 23, 2023 – July 26, 2023 · AAPM 65th Annual Meeting & Exhibition

ePoster viewing · Houston, Texas, USA

presenting author · Presenter: Udbhav S. Ram

Presented

PO-GePV-D-39

Viewing period for one ePoster

Event record and sources

Using Deep Learning models to auto-contour abdominal organs in CT scans

March 23, 2023 · Physics Undergraduate Conference (PhUnC)

Oral presentation · Western University, London, Ontario

Presenter · Presenter: Udbhav Ram

Presented

Second place overall talk, PhUnC 2023 (speaker-reported)

Event record and sources

Abdominal Organs Auto-Segmentation to Reduce Patient-Wait Times in Radiation Oncology Treatment Starts

October 29, 2022 · Canadian Undergraduate Physics Conference (CUPC 2022)

Oral presentation, Biomedical Physics session · Delta Hotels Guelph Conference Centre, Group Dining Room, Guelph, Ontario

Presenting first author · Presenter: Udbhav Ram

Presented

Third place in Biomedical Physics oral presentation judging, CUPC 2022

Event record and sources

Abdominal Multi-Organ Segmentation using Deep learning to enable enhanced workflows for LMIC

August 25, 2022 · Canadian Undergraduate Medical Physics Conference

Oral presentation · Virtual

Presenter · Presenter: Udbhav Ram

Presented

Event record and sources

An augmented-reality setup to improve accuracy in surgical chronic kidney mice model

June 7, 2022 · 2022 CAP Congress

Oral undergraduate competition presentation · McMaster University, MDCL 1110, Hamilton, Ontario

Presenting author · Presenter: Udbhav Ram

Presented

Event record and sources

Using Artificial Intelligence and Computer vision to empower better clinical outcomes in the % nephrectomy surgical model

March 2022 · Hamilton Centre for Kidney Research

Invited research talk · Hamilton Centre for Kidney Research

Invited speaker · Presenter: Udbhav Ram

Listed in authored CV

Event record and sources

The quantitative analysis of gel electrophoresis data using a cloud-based smartphone application

June 9, 2021 · 2021 CAP Virtual Congress

Oral undergraduate competition presentation · Virtual (Underline Conference System)

Presenting author · Presenter: Udbhav Ram

Presented

Event record and sources

Student Outreach

McMaster SCEC co-op experience presentations | By March 2026

McMaster Science Careers and Experience Centre

Four completed talks reported by the speaker · Speaker and peer mentor

Four completed outreach talks are reported as one aggregate. Individual dates and titles are not fully resolved.

Outreach record and source

Honors & Awards

- **Science Co-op Student of the Year, Year 5** · McMaster University · 2026
- **Blue Ribbon Poster** · American Association of Physicists in Medicine · 2025
- **Global Experience Award** · McMaster University · 2025
- **Outstanding Poster Presentation** · AAPM / Society of Physics Students · 2024
- **Best Talk, 1st Prize** · Canadian Undergraduate Physics Conference · 2023
- **Academic Entrance Scholarship** · McMaster University, Faculty of Science · 2021
- **Advanced Placement Scholar with Distinction** · College Board · 2021

Selected Appointments & Leadership

UAB-McMaster Ambassador

Mary Heersink Institute for Global Health / McMaster University

- Supported visiting students and trainees, cross-institution communication, outreach and the undergraduate exchange experience.

Additional leadership roles have included President of Hindu YUVA at McMaster, Vice President of Cultural Affairs for Hindu YUVA at UAB, and Vice President, Social, of the McMaster Undergraduate Physics Society.

Community Service & Volunteer Work

Volunteer | Oct 2021-Jan 2024

Humber River Hospital, Toronto, ON

- Supported medical imaging, surgical inpatient and information services by guiding patients and visitors and restocking medical supplies.

Certifications & Training

- Ethos Adaptive Radiotherapy Course · University of Alabama at Birmingham · 2025
- Stereotactic Radiosurgery Course · University of Alabama at Birmingham · 2025
- Advanced Open Water Diver · Professional Association of Diving Instructors · 2025
- Grade C Racing License · Federation of Motor Sports Clubs of India · 2023
- Royal Conservatory of Music · Level 8 Violin and Level 9 Piano · 2019
- Yoga teacher training (RYT-200) · Yoga Alliance · 2019
- French Language Certification (DELF-B2) · 2019

Technical Skills

Programming and software: Python, machine learning, deep learning, computer vision, full-stack engineering, DevOps, MIDAS, Pi Toolbox and SBG RaceWatch.

Research and engineering: Medical physics, radiation oncology, clinical data science, molecular dynamics, genetics, modelling, LLM integration, motorsport analytics, race engineering, performance engineering and augmented reality.

Languages: English (native), Tamil (fluent), French (fluent), Spanish (intermediate).

Open-Source Experience

Beta testing and contributions involving openpilot (comma.ai), Project MONAI and Devin. Public software profile: github.com/udiram.